

Portfolio X-Ray Application Requirements

Project Overview

Build a Python-based quantitative portfolio analysis tool that implements factor models from Giuseppe Paleologo's "Advanced Portfolio Management" book. The tool analyzes equity portfolios to decompose returns into systematic (factor-driven) and idiosyncratic (stock-specific) components.

Core Functionality

1. Data Input

Portfolio CSV Format:

- Required columns: `Ticker`, `Shares`, `BuyDate`
- Optional column: `SellDate` (if blank, assumes position still held)
- Example:

```
Ticker,Shares,BuyDate,SellDate
AAPL,100,2023-01-15,
MSFT,50,2023-02-20,2023-11-10
GOOGL,75,2023-03-01,
```

Data Fetching:

- Fetch historical adjusted close prices using yfinance
- Fetch benchmark data (S&P 500 via SPY)
- Fetch Fama-French factor data (Market, Size, Value, Momentum)
- Date range: From earliest BuyDate - 1 year to present

2. Factor Model Implementation

Factors to Calculate (Fama-French 3-Factor + Momentum):

- Market (MKT):** S&P 500 returns
- Size (SMB):** Small cap minus big cap premium
- Value (HML):** High book-to-market minus low
- Momentum (MOM):** 12-month momentum factor

Factor Loading Estimation:

- Calculate beta for each stock to each factor via regression

- Use rolling 252-day (1-year) window for beta estimation
- Store in factor loadings matrix B (n stocks $\times$ 4 factors)

Return Decomposition:

- Implement master equation: $r = \alpha + B \cdot f + \varepsilon$
- Separate returns into:
 - Systematic component: $B \cdot f$ (factor-driven)
 - Idiosyncratic component: ε (stock-specific)

3. Risk Analysis

Portfolio-Level Metrics:

- Total portfolio volatility (annualized %)
- Factor risk (systematic volatility)
- Idiosyncratic risk (stock-specific volatility)
- **% Idiosyncratic Variance:** Key metric showing portfolio "purity"
 - Formula: $(\text{idio variance}) / (\text{total variance})$
 - Target: >75% for good factor-neutral portfolio

Factor Exposures:

- Dollar exposure to each factor
- Percentage exposure relative to portfolio GMV
- Marginal contribution to factor risk (MCFR) by position

Single Position Analysis:

- Contribution to total risk
- Factor exposures per position
- Idiosyncratic volatility per position

4. Performance Metrics

Risk-Adjusted Returns:

- **Sharpe Ratio:** $\text{mean}(\text{returns}) / \text{std}(\text{returns})$
 - Calculate both daily and annualized versions
 - Annualized: $\text{daily_sharpe} \times \text{sqrt}(252)$
- **Information Ratio:** $\text{mean}(\text{idio_returns}) / \text{std}(\text{idio_returns})$

- Measures pure alpha skill independent of factors

Performance Attribution:

- Total PnL breakdown:
 - Factor PnL by factor type
 - Idiosyncratic PnL
 - Time series showing cumulative contribution

Benchmark Comparison:

- User can select benchmark (default: S&P 500/SPY)
- Calculate portfolio beta to benchmark
- Show excess returns vs benchmark

5. Visualizations

Required Charts:

1. Factor Exposure Bar Chart

- X-axis: Factor names (Market, Size, Value, Momentum)
- Y-axis: Dollar exposure
- Show as % of portfolio GMV

2. Performance Attribution Stacked Area Chart

- X-axis: Time
- Y-axis: Cumulative PnL
- Stacked areas: Market PnL, Size PnL, Value PnL, Momentum PnL, Idio PnL

3. Risk Decomposition Pie Chart

- Slices: Factor variance %, Idio variance %

4. Cumulative Returns Comparison

- Portfolio vs Benchmark over time
- Show drawdowns

Chart Requirements:

- Use matplotlib
- Clean, professional styling
- Export as PNG files

- Display in interactive window when run

6. Output & Export

Excel Export (multi-sheet workbook):

- **Summary Sheet:** Key metrics table
 - Sharpe Ratio, Information Ratio
 - Total/Factor/Idio volatility
 - % Idio variance
 - Factor exposures
- **Holdings Sheet:** Position-level details
 - Ticker, Shares, Dates
 - Current market value
 - Factor loadings (betas)
 - Contribution to portfolio risk
- **Returns Sheet:** Time series
 - Daily total returns
 - Factor returns
 - Idiosyncratic returns
- **Attribution Sheet:** Performance breakdown
 - PnL by factor
 - Cumulative attribution

Console Output:

- Print summary statistics to terminal
- Progress indicators during data fetch
- Error messages for failed ticker fetches

Technical Architecture

File Structure

```
portfolio-xray/  
├── data_loader.py    # CSV parsing, price fetching  
├── factors.py        # Factor calculations and loading estimation  
├── portfolio.py      # Main Portfolio class with risk decomposition  
└── metrics.py        # Sharpe, IR, performance calculations
```

```
|— visualizer.py      # Chart generation
|— main.py           # CLI interface
|— requirements.txt   # Python dependencies
|— example_portfolio.csv # Sample data
|— README.md         # Usage instructions
```

Dependencies

```
pandas >= 2.0.0
numpy >= 1.24.0
yfinance >= 0.2.0
matplotlib >= 3.7.0
openpyxl >= 3.1.0
scipy >= 1.11.0
```

Main Class: Portfolio

Constructor:

```
python

Portfolio(csv_path, start_date=None, end_date=None, benchmark='SPY')
```

Key Methods:

- `load_data()`: Load portfolio and fetch prices
- `calculate_factor_exposures()`: Compute B matrix
- `decompose_returns()`: Separate systematic vs idio
- `calculate_metrics()`: Compute Sharpe, IR, etc.
- `generate_report()`: Create visualizations
- `export_to_excel(filename)`: Save results

Usage Example

```
python
```

```
from portfolio import Portfolio
```

```
# Initialize
```

```
pf = Portfolio('my_holdings.csv', benchmark='SPY')
```

```
# Run analysis
```

```
pf.load_data()
```

```
pf.calculate_factor_exposures()
```

```
pf.decompose_returns()
```

```
results = pf.calculate_metrics()
```

```
# Generate outputs
```

```
pf.generate_report() # Creates charts
```

```
pf.export_to_excel('portfolio_analysis.xlsx')
```

```
# Print summary
```

```
print(f"Sharpe Ratio: {results['sharpe']:.2f}")
```

```
print(f"Information Ratio: {results['ir']:.2f}")
```

```
print(f"% Idio Variance: {results['pct_idio_var']:.1f}%")
```

Command Line Interface

```
bash
```

```
# Basic usage
```

```
python main.py my_portfolio.csv
```

```
# With custom date range
```

```
python main.py my_portfolio.csv --start 2023-01-01 --end 2024-12-31
```

```
# With custom benchmark
```

```
python main.py my_portfolio.csv --benchmark QQQ
```

```
# Export only (no charts)
```

```
python main.py my_portfolio.csv --export-only
```

```
# Verbose output
```

```
python main.py my_portfolio.csv --verbose
```

Error Handling Requirements

Must handle gracefully:

- Missing ticker data (stock delisted, API failure)

- Invalid CSV format
- Date parsing errors
- Insufficient price history
- Network timeouts
- Missing factor data

For each error:

- Print clear error message
- Continue with available data where possible
- Log failed tickers
- Don't crash the entire analysis

Data Sources

Stock Prices:

- Primary: Yahoo Finance via yfinance
- Adjusted close prices (accounts for splits/dividends)

Fama-French Factors:

- Option 1: Download from Kenneth French data library
 - URL: https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html
- Option 2: Construct from ETF proxies:
 - Market: SPY
 - Size: IWM (small) - SPY (large)
 - Value: IWD (value) - IWG (growth)
 - Momentum: Construct from stock universe

Performance Requirements

- Handle portfolios up to 200 positions
- Process 2+ years of daily data
- Complete analysis in <60 seconds
- Memory efficient (work with typical laptop RAM)

Future Enhancements (Not Required Now)

Phase 2 possibilities:

- Custom factor definitions
- Optimization module (position sizing)
- Stop-loss simulation
- Basket tracking tool
- Web interface
- Real-time monitoring
- Multi-portfolio comparison
- Transaction cost analysis

Success Criteria

The tool is successful if it:

1. ☒ Correctly loads portfolio CSV
2. ☒ Fetches all available price data
3. ☒ Calculates factor exposures accurately per book formulas
4. ☒ Decomposes returns into systematic vs idiosyncratic
5. ☒ Computes Sharpe and Information Ratios correctly
6. ☒ Shows % idio variance metric
7. ☒ Generates clear visualizations
8. ☒ Exports comprehensive Excel report
9. ☒ Handles errors gracefully
10. ☒ Runs end-to-end without crashes

Testing Plan

Test with sample portfolio:

- 10-20 positions
- Mix of large/small cap
- Mix of sectors
- Date range: 2022-2024
- Include one delisted ticker (test error handling)

Validate outputs:

- Check Sharpe ratios are reasonable (0.5-3.0 range)
- Verify % idio variance is between 0-100%
- Confirm factor exposures sum correctly
- Compare benchmark beta to expected value
- Ensure charts display correctly